

TA Class 3

The Household Block: Labour Supply, Saving, and General Equilibrium

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Macroeconomics 1 - MPE

12/09/2026

Today's Class

Three pieces we already have are one object

- Consumption–saving over many periods (TA Class 2) and the labour supply decision (PSET 4) are **one household problem**
- Firms and market clearing close it: the **household block** inside general equilibrium (Lecture 7, PSET 5)
- How a shock reaches consumption: its **direct** effect and the **indirect** effects that run through wages and interest rates
- Where heterogeneity and **labour market frictions** come in

The Household Block

Three Problems You Have Already Solved

Problem	Chooses	Takes as given	Optimality condition
Consumption–saving TA Class 2	c_t, a_{t+1}	r_t , income path	$u'(c_t) = \beta(1 + r_{t+1}) u'(c_{t+1})$
Labour supply PSET 4	c, l	w, τ, T	$v'(l)/u'(c) = w(1 - \tau)$
Firms and equilibrium Lecture 7, PSET 5	K_t, L_t	prices, technology	$w_t = F_L, \quad r_t^K = F_K, \quad 1 + r_t = r_t^K + 1 - \delta$

Lecture 7 already put them together

$$\frac{v'(l_t)}{u'(c_t)} = F_L(K_t, L_t), \quad \frac{u'(c_t)}{u'(c_{t+1})} = \beta(1 + F_K(K_{t+1}, L_{t+1}) - \delta)$$

Today: why the first two rows are **one** problem, and what exactly general equilibrium closes.

Consumption–Saving Meets Labour Supply

Consumption–saving

TA Class 2

$$\begin{aligned} \max_{\{c_t, a_{t+1}\}} \quad & \sum_t \beta^t u(c_t) \\ \text{s.t.} \quad & c_t + a_{t+1} = (1 + r_t) a_t + y_t, \quad y_t \text{ given} \end{aligned}$$

Labour supply

PSET 4

$$\begin{aligned} \max_{c, l} \quad & u(c) + v(l) \\ \text{s.t.} \quad & c = w(1 - l) + T, \quad \text{one period} \end{aligned}$$



The household problem

$$\max_{\{c_t, l_t, a_{t+1}\}} \sum_t \beta^t [u(c_t) + v(l_t)] \quad \text{s.t.} \quad c_t + a_{t+1} = (1 + r_t) a_t + w_t(1 - l_t) + T_t$$

- **Income is now a choice:** the given path y_t becomes $w_t(1 - l_t) + T_t$
- **One budget:** hours and assets are two ways to finance consumption, so work and saving are decided jointly
- **Same prices, same wealth:** r_t and w_t enter both decisions

Two Margins, One Marginal Value of Wealth

With a multiplier λ_t on each period's budget,

$$\mathcal{L} = \sum_t \beta^t \left\{ u(c_t) + v(l_t) + \lambda_t [(1+r_t) a_t + w_t(1-l_t) + T_t - c_t - a_{t+1}] \right\}$$

$$[c_t]: u'(c_t) = \lambda_t \quad [l_t]: v'(l_t) = \lambda_t w_t \quad [a_{t+1}]: \lambda_t = \beta(1+r_{t+1}) \lambda_{t+1}$$

Within a date: labour supply

$$\frac{v'(l_t)}{u'(c_t)} = w_t$$

PSET 4's condition, now holding in every period

Across dates: consumption–saving

$$u'(c_t) = \beta(1+r_{t+1}) u'(c_{t+1})$$

TA Class 2's Euler equation, now with income chosen

λ_t links the two margins: whatever changes the value of wealth (a transfer, the interest rate, a permanent wage change) moves consumption and hours together.

Saving and Investment: Who Owns the Capital?

(A) Households own K and rent it to firms

$$c_t + K_{t+1} = w_t(1 - l_t) + (1 + r_t^K - \delta) K_t$$

- Firms rent capital each period:
 $r_t^K = F_K(K_t, L_t)$
- Saving *is* investment: the household buys K_{t+1} itself
- Return on saving: $1 + r_{t+1}^K - \delta$

(B) Firms own K and borrow from households

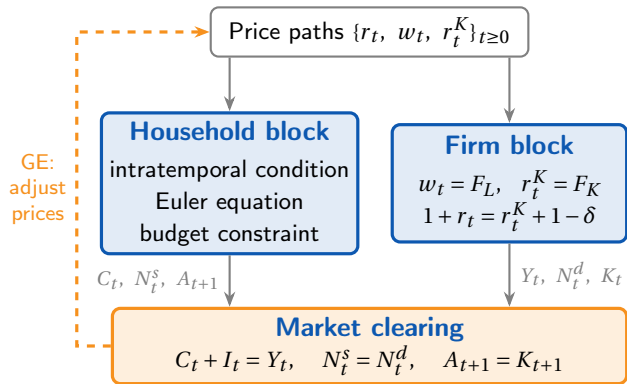
$$c_t + a_{t+1} = w_t(1 - l_t) + (1 + r_t) a_t$$

- Firms borrow to buy K_{t+1} : each unit costs $1 + r_{t+1}$ and returns $F_K(K_{t+1}, L_{t+1}) + 1 - \delta$
- Saving finances investment through the asset market: $a_{t+1} = K_{t+1}$
- Return on saving: $1 + r_{t+1}$

Same allocation

No arbitrage: $1 + r_{t+1} = F_K(K_{t+1}, L_{t+1}) + 1 - \delta = r_{t+1}^K + 1 - \delta$. Euler equation and resource constraint are identical, so who holds the capital does not matter without financial frictions (Modigliani and Miller, 1958). With frictions, it does (Bernanke and Gertler, 1989; Kiyotaki and Moore, 1997).

The Household Block



- A **block** is a function: *paths* of prices in, *paths* of quantities out. General equilibrium is the set of price paths at which the blocks agree (Lecture 7)
- RBC, New Keynesian and HANK models draw this same picture. They differ in what sits *inside* the boxes
- Household saving meets investment in the asset market, $A_{t+1} = K_{t+1}$, whoever holds the capital

Direct and Indirect Effects

How a Shock Reaches Consumption

With $u(c) = \log c$ and $v(l) = \psi \log l$, the household block solves in closed form

Euler equation + intratemporal condition + budget constraint

$$c_0 = \frac{1-\beta}{1+\psi} \left[(1+r_0) a_0 + \sum_{t \geq 0} q_t (w_t + T_t) \right], \quad q_t = \prod_{s=1}^t \frac{1}{1+r_s}$$
$$\frac{c_{t+1}}{c_t} = \beta(1+r_{t+1}), \quad l_t = \frac{\psi c_t}{w_t}$$

Transfers T_t

Raise wealth at given prices; the Euler equation spreads the gain, part of it as leisure

Wages w_t

Raise the value of time: more labour income now and later, more hours at a given c_t

Interest rates r_t

Discount future income through q_t and set the slope of consumption

- **Direct effect:** the shock itself, ΔT or Δr , with all other prices fixed
- **Indirect effect:** hours and saving change, so wages and rates move in equilibrium and re-enter all three equations (Kaplan, Moll, and Violante, 2018)

Euler fixes only the slope c_{t+1}/c_t : the level depends on the whole path of wages and rates

Tracing Two Shocks Through the Block

The household block is the same in both; the rest of the economy decides how wages and rates respond

One-off transfer $\Delta T > 0$

flexible prices, Lecture 7

Direct (prices fixed)

- Budget: wealth rises by ΔT
- Euler: spread evenly, a small permanent rise in c
- Intratemporal: richer, so fewer hours for good

Indirect (general equilibrium)

- $K \uparrow$, hours $\downarrow$: $w = F_L \uparrow$, $r = F_K - \delta \downarrow$
- Euler: lower r pulls c and leisure to today
- Intratemporal: higher w pulls hours back up

c jumps by far more, then returns to steady state

Interest-rate cut $\Delta r < 0$

prices adjust slowly

Direct (wages fixed)

- Euler: saving pays less, so c moves toward today
- Budget: savers earn less interest, pushing back

Indirect (general equilibrium)

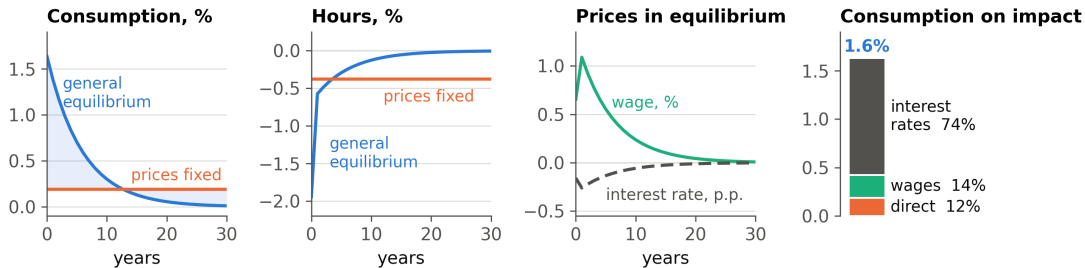
- Firms produce what is demanded: hours $\uparrow$
- Intratemporal: $w \uparrow$ so households supply them
- Budget: labour income $\uparrow$ lifts the path Euler tilts

c rises via the rate and the income spending creates

Moving the real rate needs prices that adjust slowly (a departure below). A transfer repaid by future taxes does nothing: Ricardian equivalence.

A One-Off Transfer, With and Without General Equilibrium

Same household, same shock, same Euler equation: only the prices differ

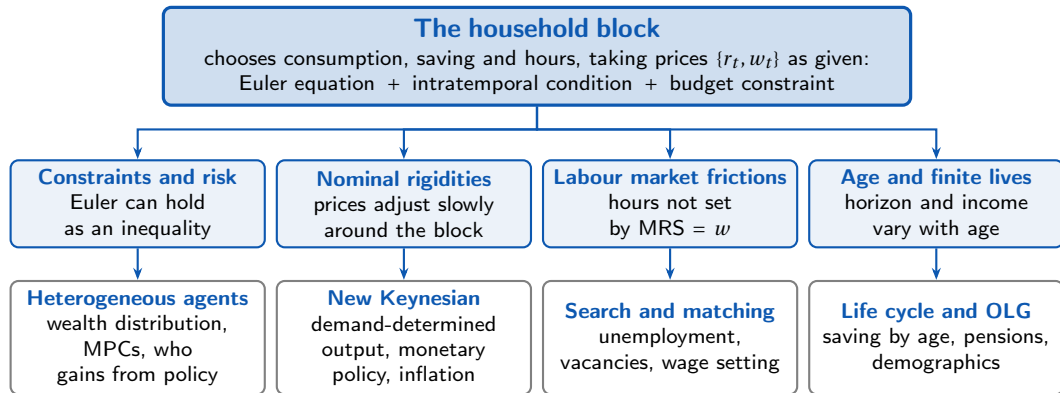


- **Prices fixed:** with $\beta(1+r) = 1$, Euler spreads ΔT evenly: consumption +0.19%, hours -0.38%, forever
- **General equilibrium:** more capital and fewer hours raise w (+1.1% in year 1) and lower r (-0.26 p.p.). Consumption jumps 1.6%, **8.6 times** the direct effect; hours fall 1.9%, since $\% \Delta l = \% \Delta c - \% \Delta w \approx 1.6 - 0.7$
- **Channels on impact:** 12% direct, 14% wages, 74% rates. Euler matters most, but at the equilibrium rate

Annual: $\beta = 0.96$, $\alpha = 1/3$, $\delta = 0.08$, hours $1/3$ ($\psi = 1.71$); $\Delta T = 10\%$ of output at $t = 0$, from abroad. Split adds equilibrium wages, then rates, to the fixed-price household. Script: windfall_ge_figures.R

One Household Block, Many Models

Keep the core, change one piece: each departure opens a different kind of analysis



Departures stack

HANK combines the first two branches: heterogeneous households inside a New Keynesian economy. Add the third and the risk of losing a job enters the household problem itself.

Heterogeneity in the Household Block: Where to Read Next

Ingredient	What it changes	Start with
Borrowing constraints and liquidity	Euler becomes an inequality; hand-to-mouth households spend what they get	Deaton (1991); Kaplan and Violante (2014); Kaplan, Violante, and Weidner (2014)
Nominal rigidities	demand sets output, so MPCs drive transmission; unequal exposure to inflation	Bilbiie (2008); McKay, Nakamura, and Steinsson (2016); Kaplan, Moll, and Violante (2018); Auclert (2019)
Income and wealth risk	precautionary saving, buffer stocks; MPCs fall with wealth	Huggett (1993); Aiyagari (1994); Carroll (1997); Krusell and Smith (1998); Blundell, Pistaferri, and Preston (2008)
Age and returns	life-cycle saving; persistent differences in returns	Gourinchas and Parker (2002); Fagereng, Guiso, Malacrino, and Pistaferri (2020)
Unemployment risk	job loss, lasting earnings losses, countercyclical risk	Merz (1995); Andolfatto (1996); Krusell, Mukoyama, and Şahin (2010); Ravn and Sterk (2017); Den Haan, Rendahl, and Riegler (2018)

Surveys: Krueger, Mitman, and Perri (2016); Kaplan and Violante (2018). Full references in the appendix.

What changes in the household block when there is involuntary unemployment?

Labour Market Frictions

What Changes When Finding a Job Takes Time

	Frictionless PSET 4, Q1	Search and matching DMP, PSET 4, Q2
Margin of adjustment	hours of every worker	employed vs. unemployed
Who sets employment	the household: $v'(l)/u'(c) = w$	household and firm (jointly)
Wage setting:	Market clearing: $w = \text{MRS} = F_L$	Bargaining over match surplus
Unemployment	Only voluntary	$u = s/(s + f(\theta))$ in steady state
Labour income risk	none, for a given w	job loss (s), slow re-employment (f)

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